

Software Requirements Engineering

SE 311

Has a SW shortage ever frustrated you?

Introduction

- Software defines our lives



Self-driving Vehicles



Unmanned Aerial Vehicles



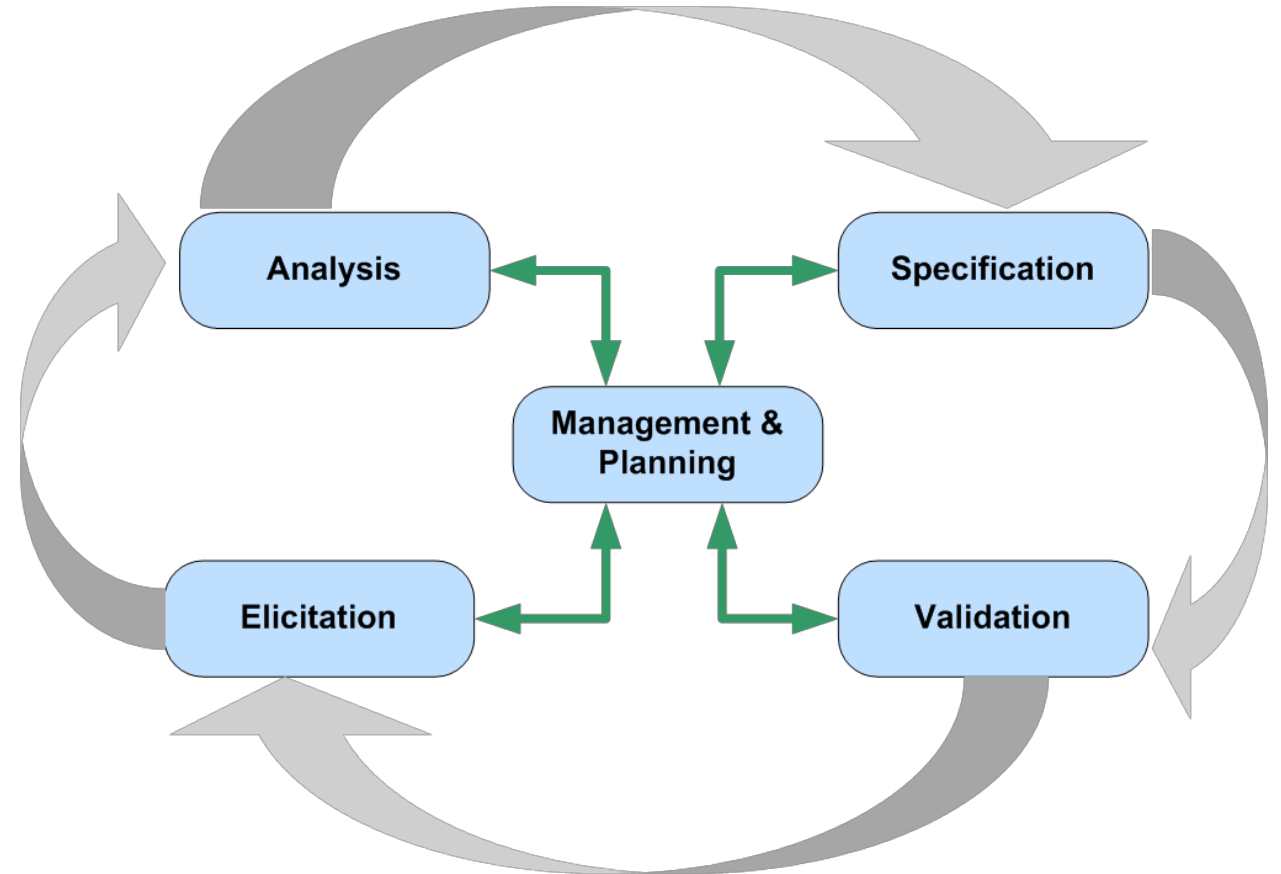
Medical Sensors & Actuators



Robots

Requirements Engineering (RE) Process

- A software system is assessed in terms of the extent to which it satisfies the purpose for which it was intended.
- RE is the process of discovering that purpose by defining a set of software requirements to delineate the constraints and demands of the system



Given that

- The **size** of the worldwide software industry is measured in billions in an annual bases
- **Errors** introduced during requirements activities account for 40 to 50 percent of all defects found in SW products
- The hardest single part of conceptually defining a software system is the task of **defining the software requirements**.
- “no other part of the work so cripples the resulting system if done wrong” and “no other part is more difficult to rectify later” (Brooks, 1987)

Course Objectives

- CLO1: Recognize the different types of requirements and importance of software requirement engineering.
- CLO2: Identify multiple techniques to elicit requirements from stakeholders, choosing from among alternative methods as appropriate for different situations.
- CLO3: Analyze documenting requirements for high-level software requirements engineering.
- CLO4: Design and develop models for requirement documentation using goals, use cases, UML class and activity diagrams and state charts.
- CLO5: Evaluate the software requirements using validation and negotiation techniques regarding priorities and scope with the stakeholders.
- CLO6: Measure the documented requirements using quality criteria techniques for accuracy, unambiguity, consistent and completeness of the requirements.
- CLO7: Demonstrate the management of requirements as they change over time. (e.g., over multiple releases and over a product line)

Class Project

- **Identify** a *real* problem in any domain of your choice where you think a SW solution is needed to solve the problem
- **Identify** *real* problem stakeholders
- **Conduct** what you've learned about RE in order to capture stakeholders needs and produce a **Software Requirements Specification** (SRS) for your proposed solution

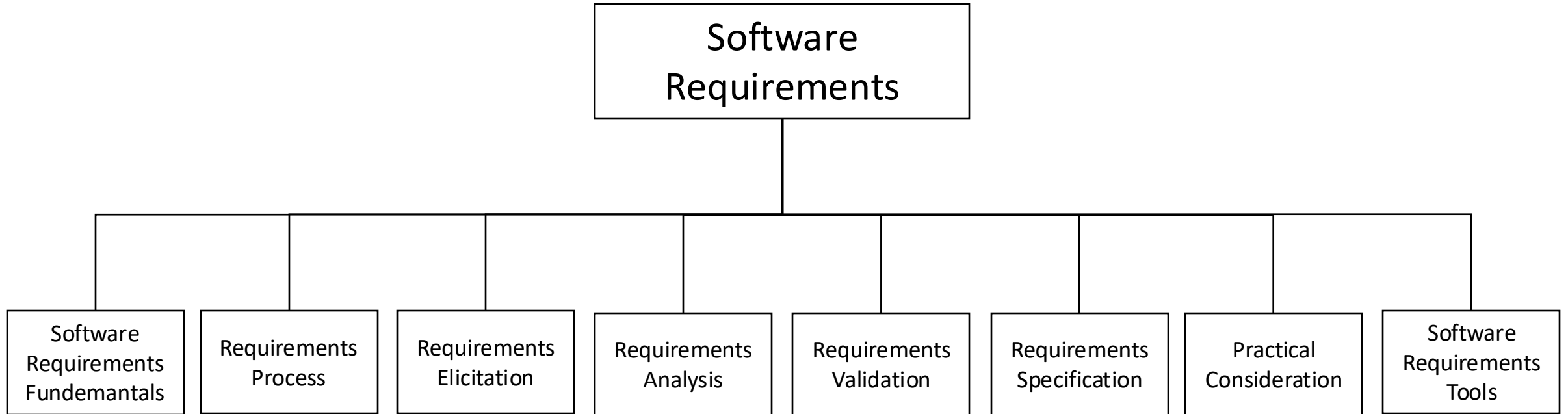
Class Project

- Teams of 3-4 students
- Select a team leader to coordinate communication and deliver submissions
- Four deliverables:

First deliverable	Second deliverable	Third deliverable	Fourth deliverable
• Vision & scope document	• Elicitation notes and evidence of elicitation process • A list of prioritized user requirements	• SRS that is approved by stakeholders • A prototype of the SW	• Presentation

Software Requirements Fundamentals

Software Requirements Knowledge Area

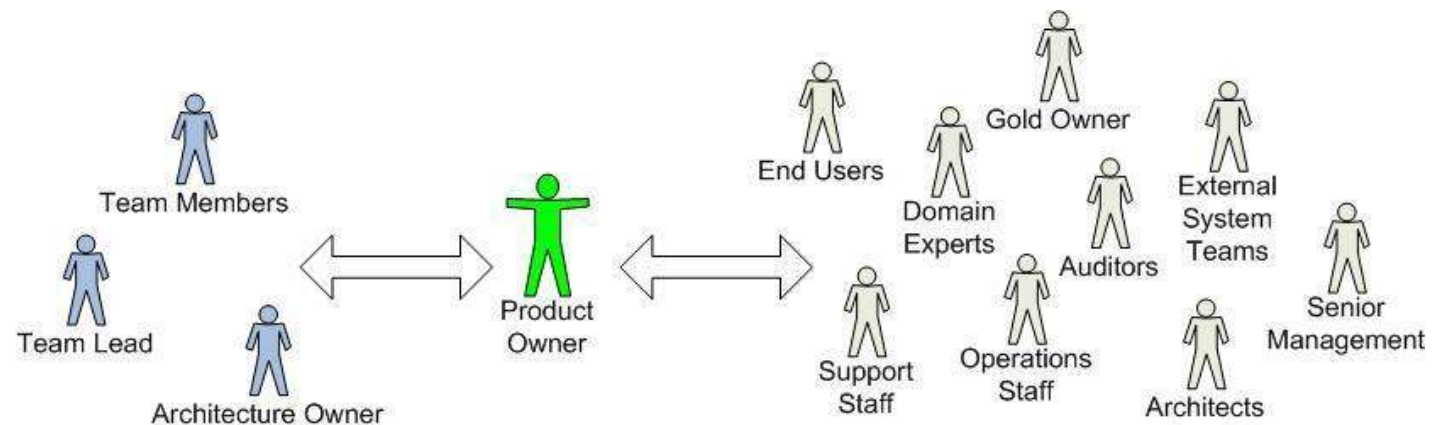


What is a *software requirement*?

- A statement which translates or expresses a need and its associated constraints and conditions (IEEE/ISO/IEC 29148-2018)
- At its most basic, a software requirement is a property that must be exhibited by something in order to solve some problem in the real world (SWEBOK V3)

What are *software requirements*?

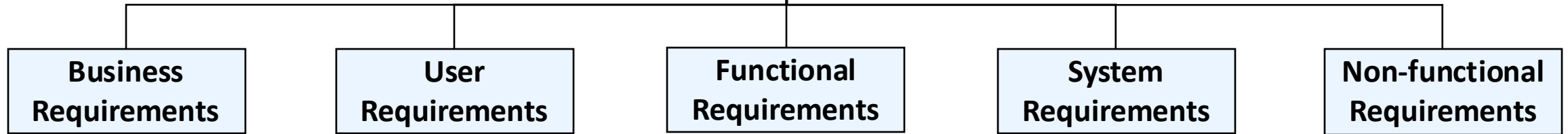
- The set of requirements form a complex matrix of the needs of many stakeholders who are spread across different levels of the organization and who are somehow linked to the problem



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Types of Requirements

Software Requirements



Definitions

Describe why the organization is implementing the system.

Describe goals or tasks the users must be able to perform with the product that will provide value to someone.

Specify the behaviors the product will exhibit under specific conditions

Describe the requirements for a product that is composed of multiple components or subsystems

Describe how *well* the products carries out its tasks.

- Quality attributes
- External interfaces
- Constraints

Examples

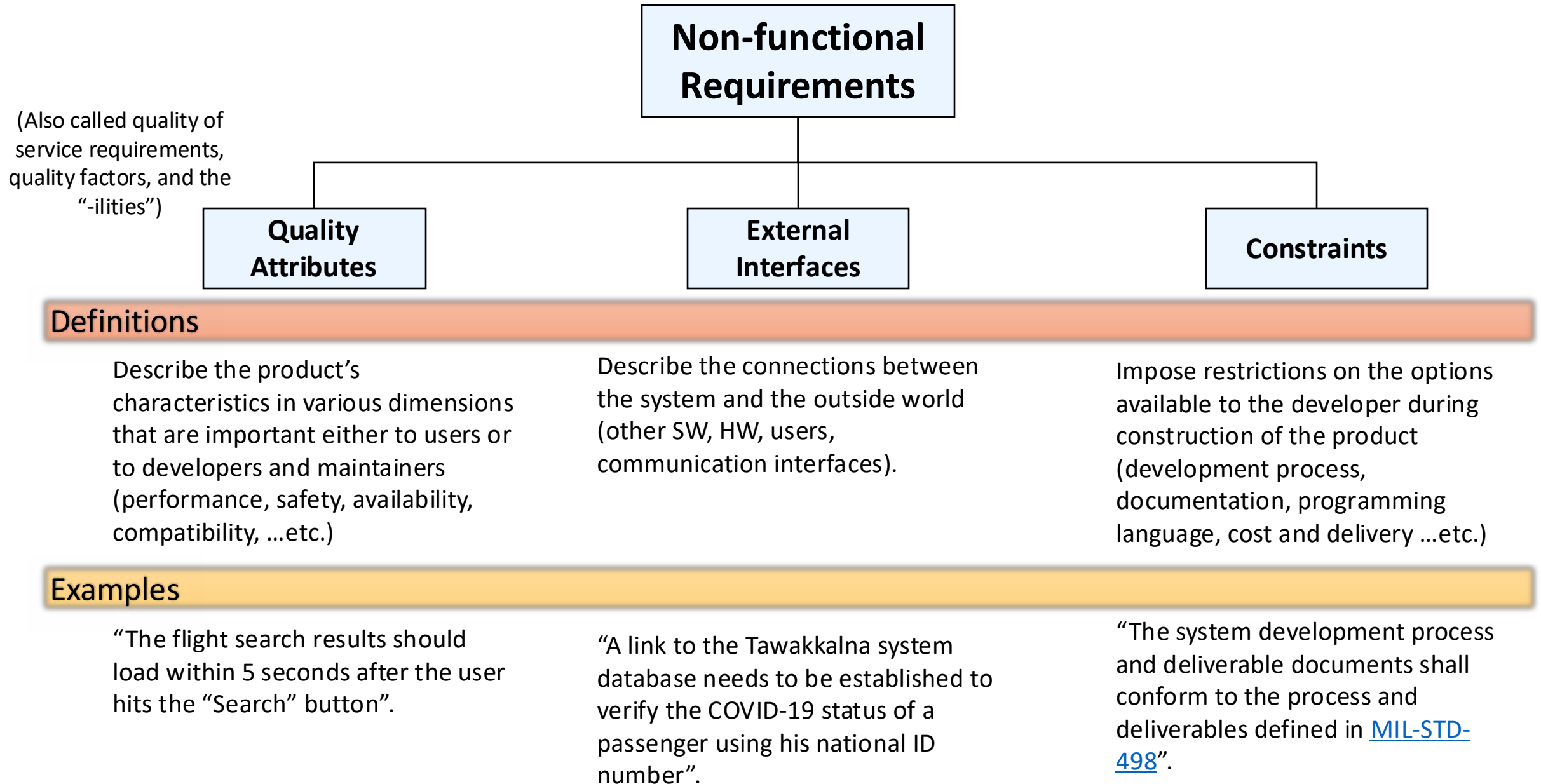
"Airline wants to reduce airport counter staff costs by 25 percent."

"User needs to be able to check-in for flight using airline's website."

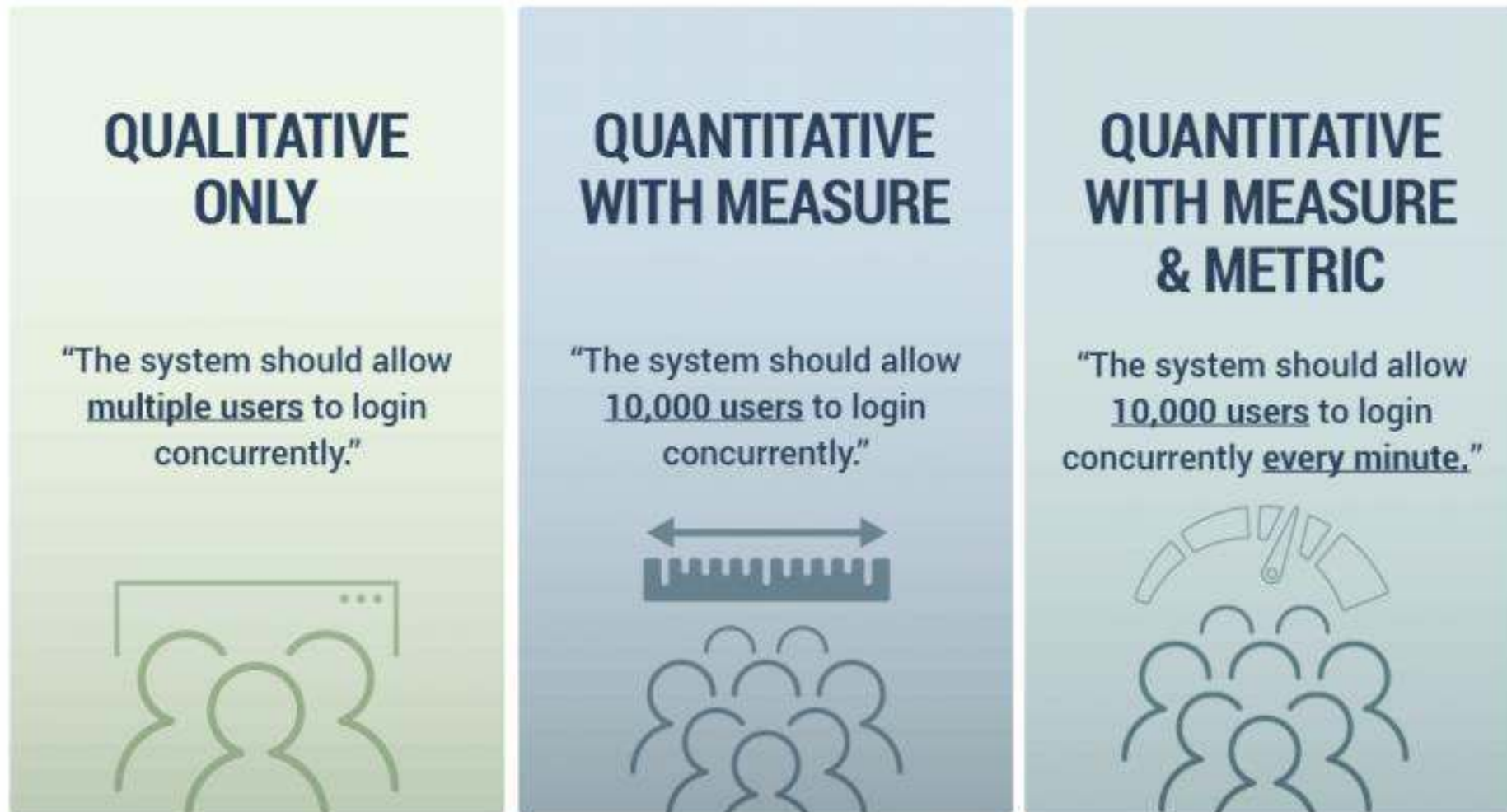
"If the Passenger's profile does not indicate a seating preference, the reservation system shall assign a seat."

"The airport check-in kiosks need a barcode reader to scan a passport."

"The airline's website needs to be available 98% of the time OR the app needs to be iOS compatible."



Expression of Non-functional Requirements



Quality requirements

Important primarily to users

- Availability
- Efficiency
- Flexibility
- Integrity

- Interoperability
- Reliability
- Robustness
- Usability

Important primarily to developers

- Maintainability
- Portability
- Reusability
- Testability

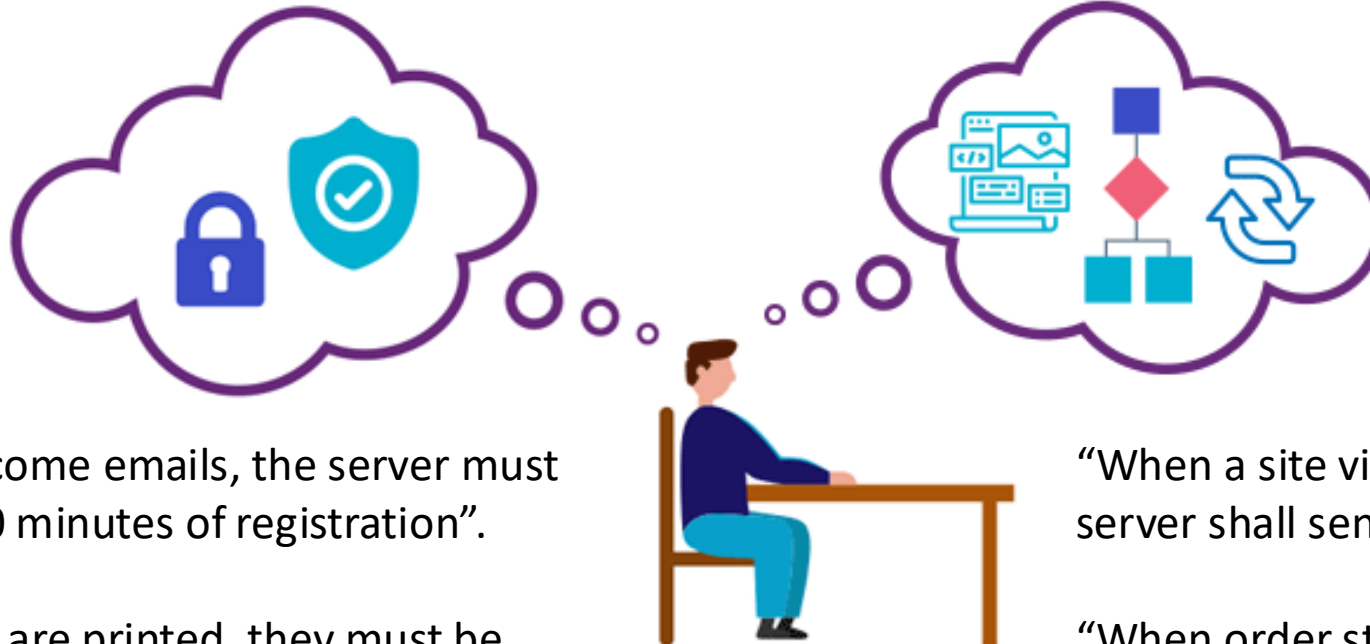
- Product **properties**
- **How** should the system do?
- Helps verify **performance**.
- **More difficult** to capture.

- Product **features**
- **What** should the system do?
- Helps verify **functionality**.
- **Easier** to capture.

Non-Functional Requirements

VS.

Functional Requirements



“When sending welcome emails, the server must send them within 10 minutes of registration”.

“When packing slips are printed, they must be on both sides of 5” x 8” sheets of white paper”.

“When a site visitor creates an account, the server shall send a welcome email”.

“When order status changes to fulfillment, the local printer shall print a packing slip”.

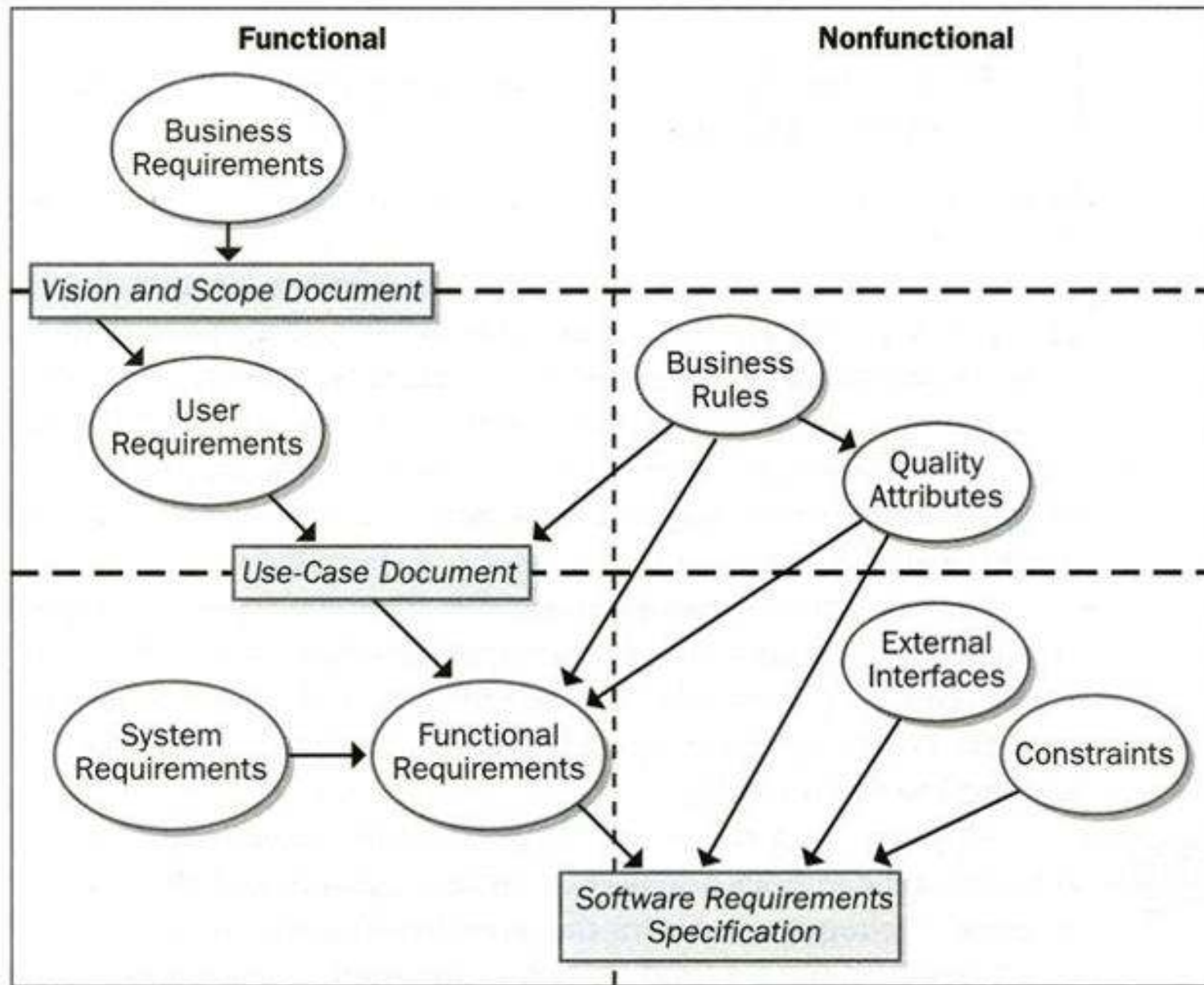
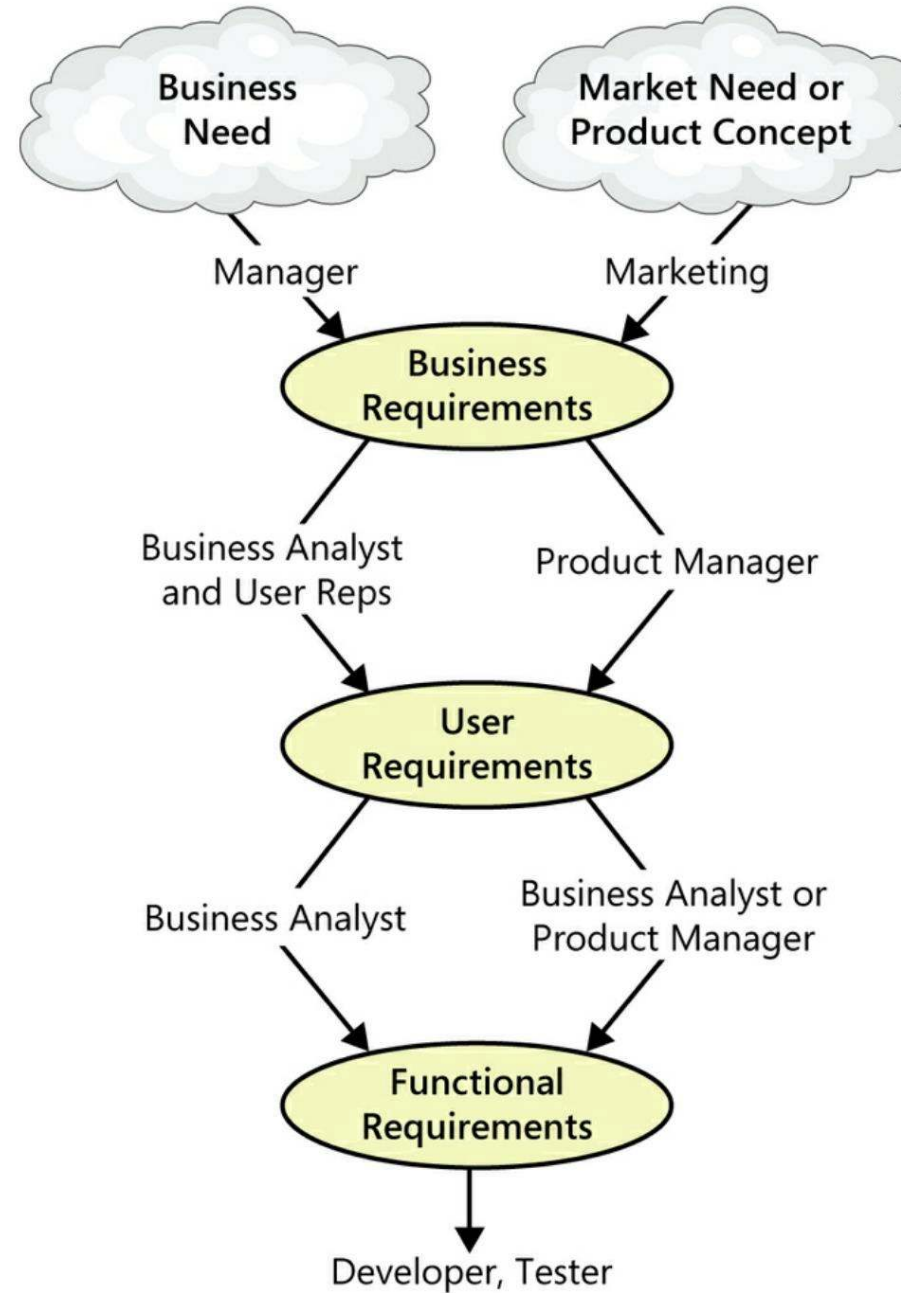


Figure 1-1 Relationship of several types of requirements information.

Corporate Roles

Commercial Roles



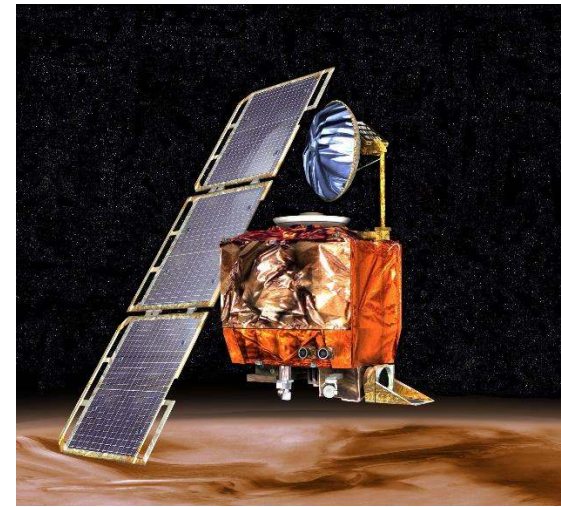
Domain Requirements

- Derived from application domain
- Describe system characteristics and features that reflect the domain
- May be new functional requirements or constraints on existing requirements
- Example: “A patient’s designated physician is the only entity allowed to retrieve a patient’s medical reports”.

Why RE?

Mars Climate Orbiter

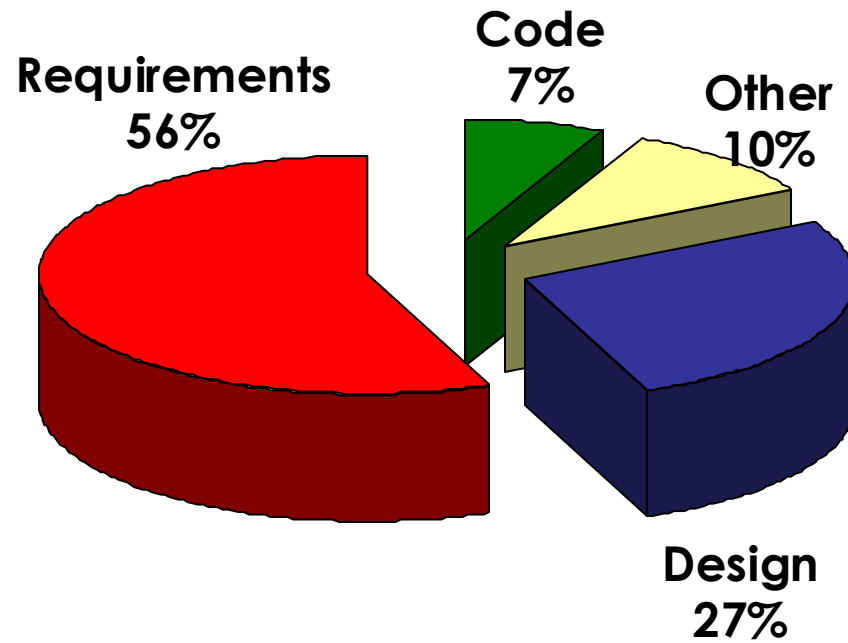
- Cost about \$235.9M
- Launched by NASA in 1998 to study the Martian climate.
- It disappeared around Mars in 1999
- Failure caused by a measurement mismatch between two SW systems: the NASA team used the metric system, while the spacecraft builder Lockheed Martin used the English system for a key function



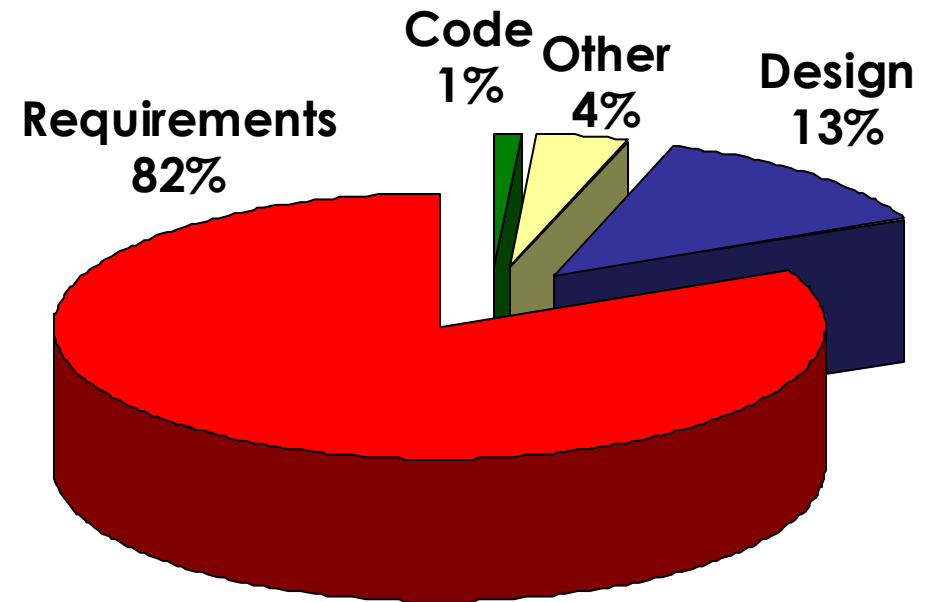
Integrated Resource Management Project (GIRES)

- GIRES (Gestion intégrée des ressources)
 - To replace >1000 existing systems in 140 organisations / departments
 - Affecting 68,000 employees
- The project started in 1998 with an estimated timeframe of 8 years and a budget of \$80 million
- Could not cope with changes to the requirements and was cancelled in 2003 after investing
- Total cost: \$400 million

Why Focus on RE?

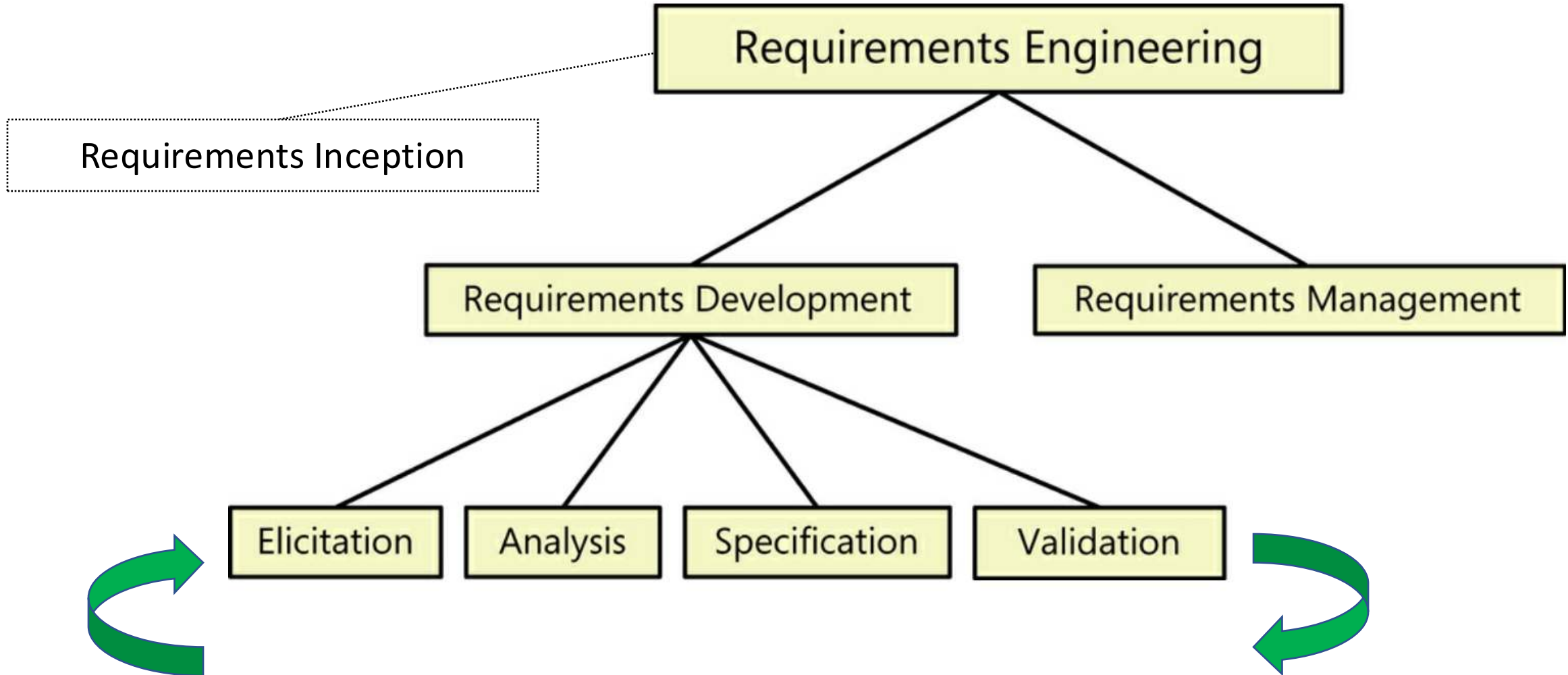


Distribution of Defects



Distribution of Effort to Fix Defects

RE Activities



RE Activity

Inception: Start the process (business need, market opportunity, great idea), business case, feasibility study, system scope, risks, etc.

Elicitation: All the tasks involved with discovering and gathering the requirements (interviews, document analysis)

Analysis: Develop a deep understating of the requirements on an individual level and as a whole (decomposition, conflicts, gaps)

Specification: Documenting these requirements in a well-organized way suitable for comprehension by their intended audiences.

Validation: Reviewing the documented requirements by internal teams and the customer to confirm that it's complete and properly represented and that it meets the customer needs.

Management: Managing the requirements beyond being agreed upon and through implementation up until the project is completed (tracing reqs, tracking status, changes – scope, assessing impact, accepting or refusing)

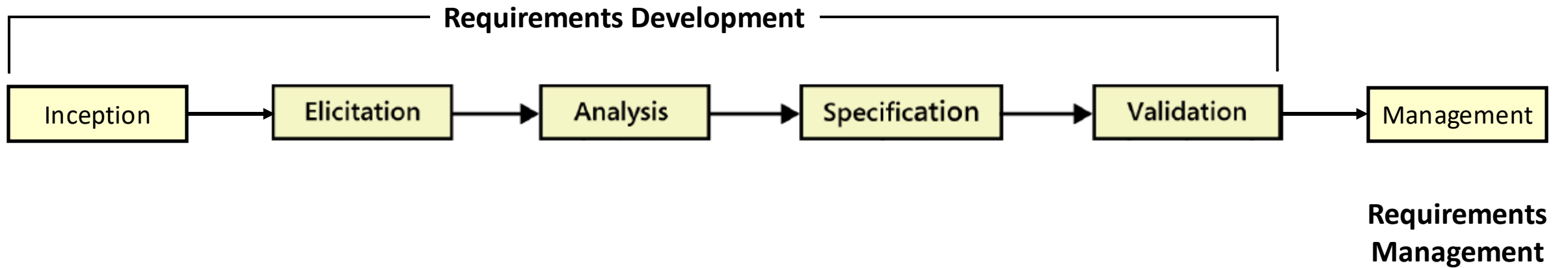
Regardless of what development life cycle is being used (waterfall, iterative, agile, ...etc.) these are the thing you need to do regarding requirements.

Iteration is key to requirements development success.

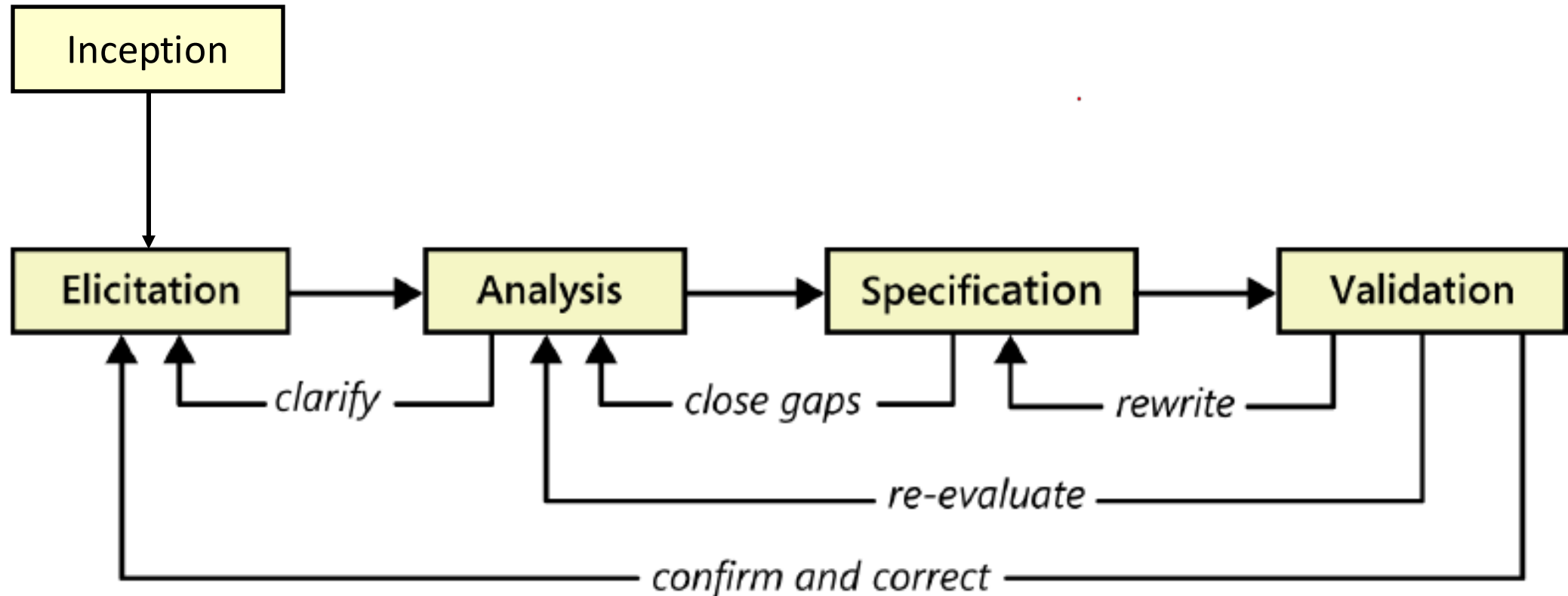
Requirements Engineering (RE) Process

- A *process* is a sequence of steps performed for a given purpose (e.g. SW development process)
- RE is a process, not a discrete activity

RE Process

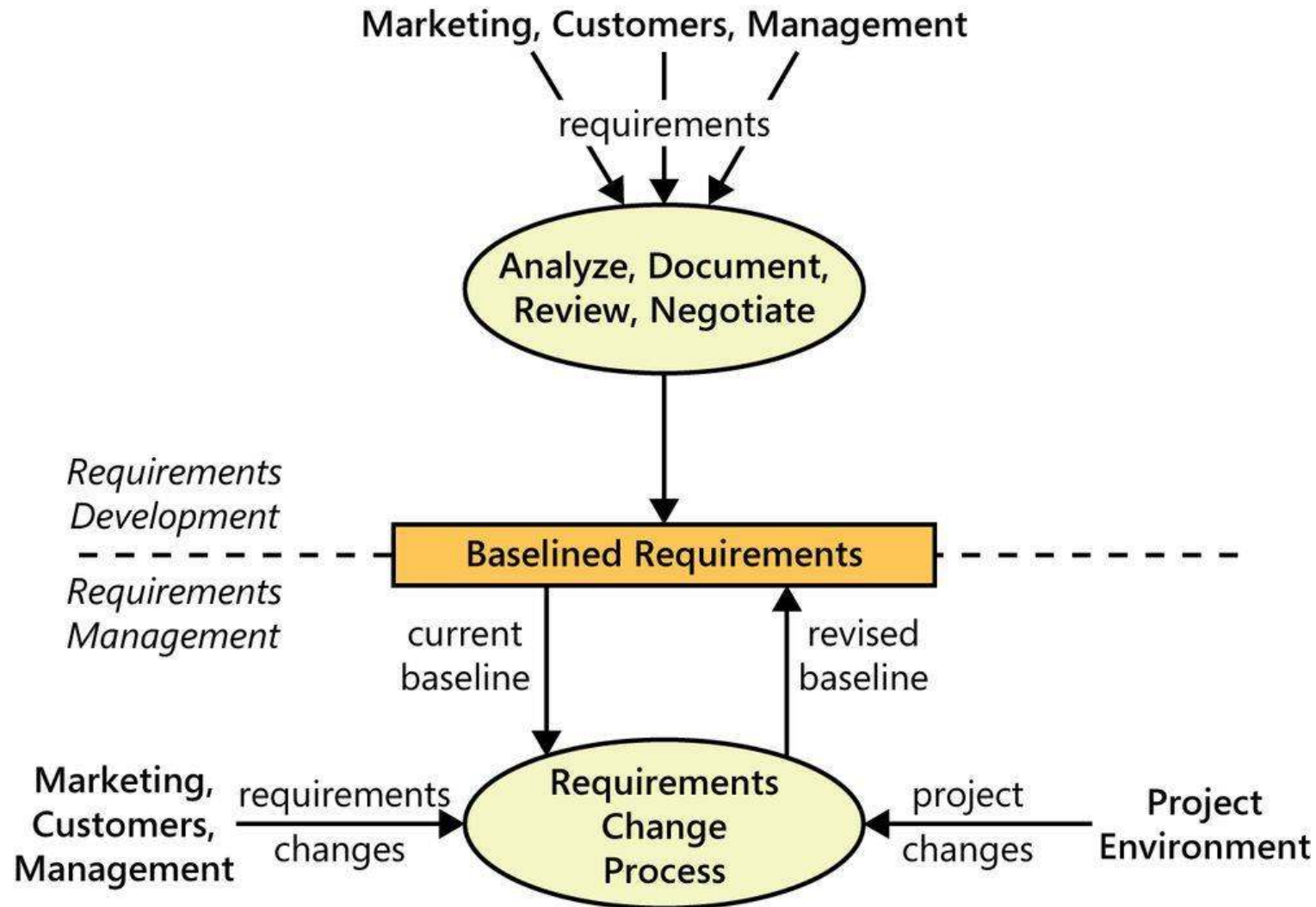


Requirements Development Process



"Progressive refinement of detail"

Boundary Between RD and RM

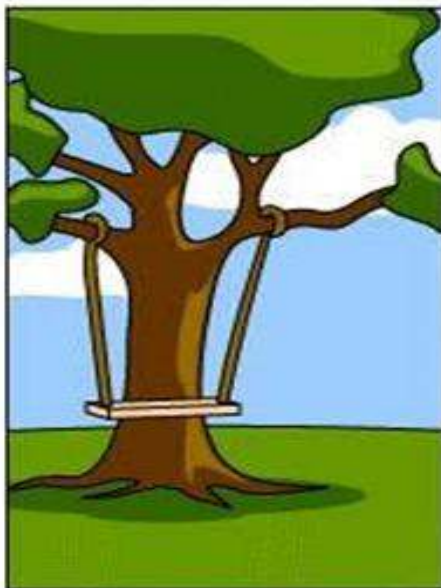


RE Challenges

- Lack of the right expertise
- Insufficient user involvement
- Overlooked stakeholders
- Unrealistic or unclear goals
- Ambiguous requirements
- Conflicting requirements
- Creeping user requirements
- Changing requirements



How the customer explained it



How the Project Leader understood it



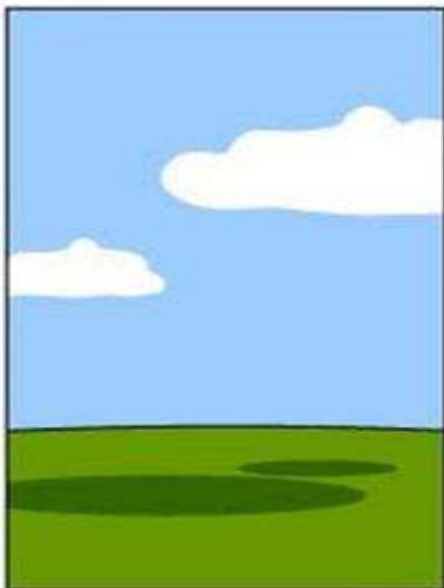
How the Analyst designed it



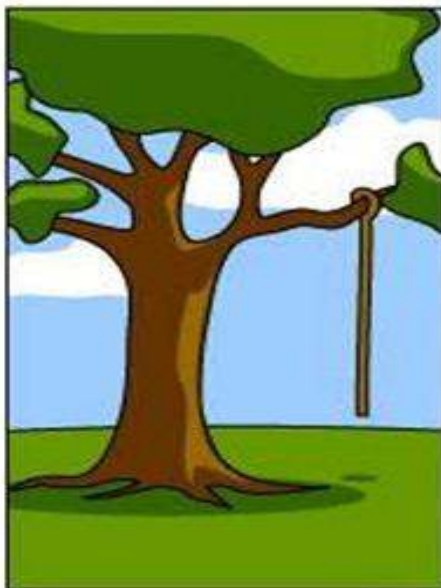
How the Programmer wrote it



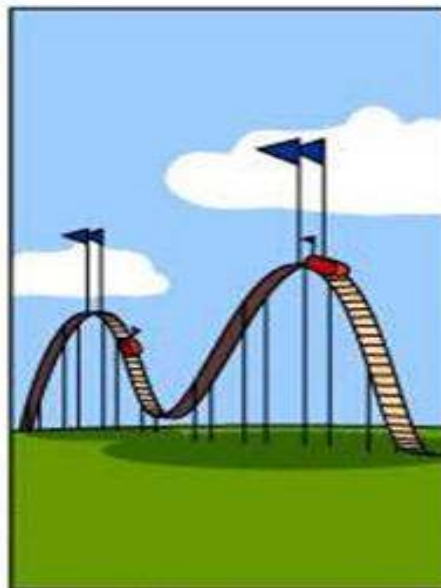
How the Business Consultant described it



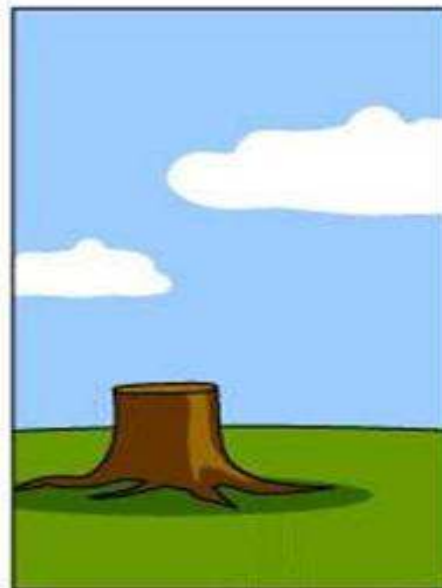
How the project was documented



What operations installed



How the customer was billed



How it was supported



What the customer really needed